Key shift

Classical math:

Human proves → paper

Foundational math:

Human defines rules → human/machine proves

Computational math:

Machine explores → human validates

What is the primary object in Category IV?

This is the key distinction.

Category Primary object

I (Substantive) Mathematical objects
II (Foundational) Structures & proofs
III (Computational) Processes & algorithms
IV (Cognitive) Conceptual representations

Philosophy of mathematics
* Mathematical logic
* Proof theory
* Type theory
* Category theory
* HoTT
* Formal verification

Model Theory

Real-world impact (non-obvious)

- Gödel's incompleteness → limits of formal AI reasoning
- Church–Turing thesis → limits of computation
- Constructivism → safe & verifiable software
- Structuralism → modern ML representations

Mathematical logic is the study of formal logic within mathematics. Major subareas include model theory, proof theory, set theory, and recursion theory (also known as computability theory). Research in mathematical logic commonly addresses the mathematical properties of formal systems of logic such as their expressive or deductive power. However, it can also include uses of logic to characterize correct mathematical reasoning or to establish foundations of mathematics.